How to Decide What's Right for Me?

When it comes to choosing your production environment, there's no right answer; one size does not fit all. Rather than give you specific advice or recommendations, instead it's important to highlight some key criteria that you'll need to think about when deciding.

Here are the trade-offs to think about:

Upfront cost vs. long-term cost. The unit cost for services generally increases with the amount they manage for you; IaaS is typically cheaper than CaaS, which is cheaper than Serverless. However, the less managed, the more upfront engineering effort is required. It's often larger, more established companies that go down the IaaS route; they have the deep pockets required, and more certainty that they'll be around long enough to benefit from the multiyear cost savings. For smaller companies, and especially startups, these upfront costs are hard to justify unless they give a key competitive advantage.

Focus and developer bandwidth. Similar to cost, can you afford the time and loss of focus spent building your platform on top of IaaS, rather than just getting something working faster with a managed service? The general wisdom is that you should focus on what makes you and your product unique—in most cases, this won't be your production environment.

Support and maintenance costs. If you use a CaaS platform, the service is managed, maintained, and improved by someone else. If things go wrong, it's most likely someone else's problem to fix. If you roll your own solution, be prepared for more support and maintenance to fall on your shoulders.

Control and flexibility. With a CaaS, you have limited control to change the fundamental features or capability of the system, whereas IaaS gives you ultimate control to build systems as you like. Consider how much control you need and along what dimensions. Some industries have very specific security or audit requirements that might necessitate a bespoke solution. However, Kubernetes gives you lots of options for architecting your software, so it may provide as much flexibility as you need.

Regional availability. If you're based in Sydney and delivering software for users in Sao Paulo, to avoid high latency, your app should probably be hosted close to the end users. Some of the managed CaaS services have limited regional availability, which may rule out certain offerings.

Level of expertise required vs. ability of team. Building your own secure cloud platform, even if leveraging some good defaults that may be provided in an